Multimedia Signal Processing - 2nd Module

June, 21 2022

1. **Question (7 pts)** Consider a discrete source X emitting the following series of symbols

$$x = \{3,\ 15,\ 9,\ 18,\ 15,\ 15,\ -3,\ 12\}.$$

You are given a quantizer $Q(\cdot)$ that applies the following quantization rule

$$Q(x) = \left\lfloor \frac{x}{\Delta} \right\rceil \cdot \Delta,$$

where $\lfloor \cdot \rceil$ rounds to the nearest integer, and $\Delta = 2$ is the quantization step.

   (a) Compute the sequence $\hat{x}_{\text{PCM}}$ obtained by encoding and decoding x using a PCM scheme.

   (b) Compute the sequence $\hat{x}_{\text{DPCM}}$ obtained by encoding and decoding x using a closed-loop DPCM scheme. The predictor simply returns the previous sample, and the first sample is encoded using PCM.

   (c) Compare the mean squared error (MSE) obtained with the PCM and with the DPCM.

   (d) Compute the Huffman code to encode $\hat{x}_{\text{PCM}}$. Estimate the needed probability mass function (p.m.f.) directly from the observations you have.

   (e) Consider the two matrices $T_1 = \begin{bmatrix} 1, & 0 \\ 0, & 1 \end{bmatrix}$ and $T_2 = \begin{bmatrix} 0, & 1 \\ 1, & 0 \end{bmatrix}$. Can they be used to apply transform coding to the sequence x? If yes, would it make any sense? There would be any difference in using one or the other?

2. **Question (4 pts)** Consider you have the bitstream representing an encoded JPEG image. With the help of a block diagram, clearly explain how to decode the image in order to go back to the pixel domain. Explain the role of each block, as well as which are the inputs and the outputs.

3. **Question (4 pts)** Explain the concept of Absolute Threshold of Hearing, and sketch it. How can this concept be used for audio perceptual coding?

4. **Questions (8 pts - each answer can be either TRUE or FALSE - Incorrect answers give you -1 points, unanswered 0 points, correct answers 1 point - Provide a brief comment when in doubt)**

   - [T] [F] Considering masking effects in audio perceptual coding, noise can only mask tones.
   - [T] [F] Transform-coding-based image compression schemes typically assume that all pixels are independent and uncorrelated in natural images.
   - [T] [F] In MP3 audio coding it is possible to change the considered window size in time.
   - [T] [F] Given two discrete sources, mutual information is directly proportional to the joint probability of the sources.
   - [T] [F] Consider the code that maps three symbols a, b and c into binary words according to the rule $C(a) = 0$, $C(b) = 1$ and $C(c) = 01$. This is a one-to-one code.
   - [T] [F] In H264 video coding, the use of prediction in intra-coded frames is allowed.
   - [T] [F] Transform coding is always a better choice than normal PCM coding in theory and practice.
   - [T] [F] It is not possible to outperform Lloyd-Max quantization considering scalar quantizers.

5. **Question (Matlab - 10pts)**

   Compare a PCM and a 2×2 block-based transform coding scheme.

   Signal Preparation

   (a) Load the default Matlab image `peppers.png` using the command `imread('peppers.png')` and display it.

   (b) Extract the green color plane. This is your signal from this point on.

   PCM

   (a) Encode the signal using a PCM scheme based on a uniform scalar quantizer. Assume that the original signal is clipped in the range $[0, 255]$ and your quantizer works with 4 bits.

   (b) Compute the entropy and the PSNR.

   Transform Coding

   (a) Encode the signal using a 2×2 block-based transform coding scheme. Use a 2×2 transform matrix of your choice (it has to be a valid matrix for transform coding). Use again a 4-bit quantizer adapted to the transformed signal dynamic.

   (b) Compute the entropy and the PSNR.

   Comparison

   (a) Show the working point of the two solutions on a rate-distortion plane, considering entropy as rate and PSNR as distortion. Comment on the meaning of this plot (i.e., what can you in principle conclude by observing it?).